Inequalities — problem 7.32

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Let $0 < p \leq a, b, c, d, e \leq q$. Show that

$$(a+b+c+d+e)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}+\frac{1}{e}\right) \leq 25 + 6\left(\sqrt{\frac{p}{q}} - \sqrt{\frac{q}{p}}\right)^2.$$

This is a problem of the US Olympiad 1977. It is a special case of a general theorem. Also, prove this more general theorem.